

SE7595

MARIAN COLLEGE KUTTIKKANAM (AUTONOMOUS)
UG Programme (Honors) Degree October 2024 Regular Examinations

First Semester

241SECUBC101 Fundamentals of Programming Using C++

Time: 2 Hours

Maximum Marks: 40

Answer one question from each bunch

BUNCH I

- I.A You are developing a program to track fuel efficiency for n vehicles. The program should perform the following tasks for each vehicle:
- Input the vehicle type (car, bike, or truck), distance traveled (in km), and fuel consumed (in liters).
 - Calculate the fuel efficiency:
 - Fuel Efficiency = Distance Traveled / Fuel Consumed.
 - Classify the fuel efficiency:
 - "Excellent" for efficiency ≥ 20 km/l, "Good" for 15–19 km/l, and "Poor" for < 15 km/l.
 - Display the vehicle type, distance, fuel consumed, fuel efficiency, and efficiency classification.

Draw a flowchart and write a step-by-step algorithm. (CO1:8 Marks)

OR

- I.B You are developing a program for an online shopping cart system for n customers. The program should perform the following tasks for each customer:
- Input the customer's name, number of items purchased, and the price of each item.
 - Calculate the total amount spent:

- Total Amount = Sum of Prices of All Items.
 - Apply a discount of 10% if the total amount exceeds Rs. 100.
 - Display the customer's name, number of items, total amount, discount (if any), and the final amount payable.
- Draw a flowchart and write a step-by-step algorithm. (CO1:8 Marks)

BUNCH II

II.A Imagine you are teaching a beginner programming class and need to demonstrate how to write and execute a simple C++ program that prints "Welcome to C++ Programming!" to the screen. In this scenario, explain the steps involved in writing, compiling, and executing this simple C++ program. What are the key components of the program, such as the #include directive, the main() function, and the cout statement? Additionally, discuss common errors that beginners might encounter when running their first C++ program, and provide guidance on how they can resolve them. (CO2:8 Marks)

OR

II.B Create a C++ program that defines a global variable length. Inside a function, declare a local variable length and assign it a value to calculate the area of a rectangle (length × width). Use the scope resolution operator to display the global and local length values after calculating the area. Also, explain the local and global scope of a variable. (CO2:8 Marks)

BUNCH III

III.A Write a C++ program that accepts 5 integers from the user and stores them in an array. The program should then find and display the largest element in the array. In your answer, explain how arrays are declared and how you used a loop to iterate through the array to find the largest element. Justify the use of the for loop for array traversal. (CO3:8 Marks)

OR

III.B Write a C++ program to count the number of vowels and consonants in a character array. (CO3:8 Marks)

BUNCH IV

IV.A Write a program to demonstrate constructor overloading in C++ using a class Rectangle.

- Define a class Rectangle with:
 - **Private Data Members:** length and width.
 - **Constructors:**
 - A default constructor that initializes length and width to 0.
 - A parameterized constructor to initialize length and width to specific values.
 - A copy constructor to create a new rectangle with the same dimensions as an existing rectangle.
 - **Public Member Functions:**
 - A function to calculate and return the area.
 - A function to display the length, width, and area of the rectangle.
- Write a program to test all three constructors by creating multiple objects and displaying their details. (CO4:8 Marks)

OR

IV.B Define a class Player with private attributes: name, age, team, and position.

Write member functions to:

- Input and display player details.
- Use a static member to track the total number of players added to the team.
- Write a program to manage the sports team and display the total number of players. (CO4:8 Marks)

BUNCH V

V.A Write a C++ program to demonstrate dynamic memory allocation and deallocation using new and delete for an array of integers.
(CO5:8 Marks)

OR

V.B Write a C++ program to define a structure Car with the fields model, year, and price. Accept input for these fields and display the details.
(CO5:8 Marks)

CO1	Analyse the problem-solving process focusing on problem analysis, design, programming languages, translators, and debugging.
CO2	Analyse C++ fundamentals including history, structure, data types, operators, and basic input/output operations.
CO3	Implement decision-making, loops, and arrays in C++, including initialization, operations, and string manipulation.
CO4	Apply OOP concepts in C++ using classes, objects, access specifiers, constructors, destructors, and static members.
CO5	Define C++ functions, including prototypes, definitions, parameters, recursion, pointers, dynamic memory, structures, and unions.